



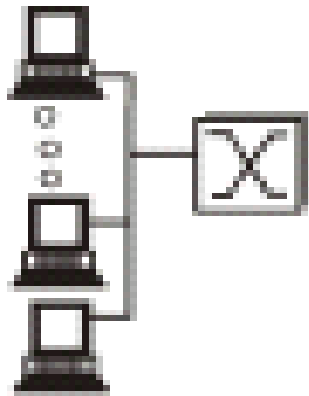
Chapter 12

Multiple Access

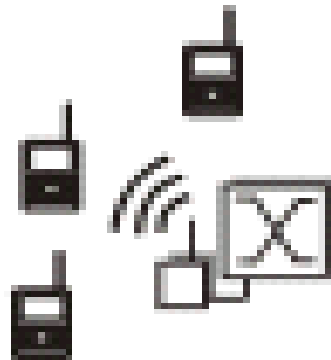
Multiple Access

- **Broadcast link** used in LAN consists of multiple sending and receiving nodes connected to or use a single shared link

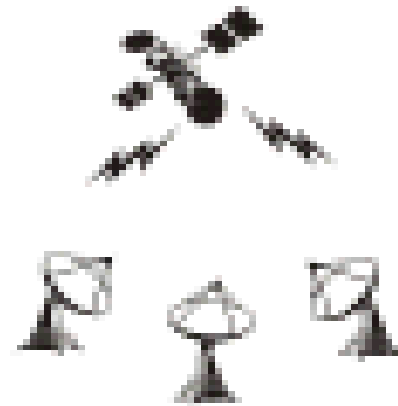
Broadcast links Examples



shared wire
(e.g. Ethernet)

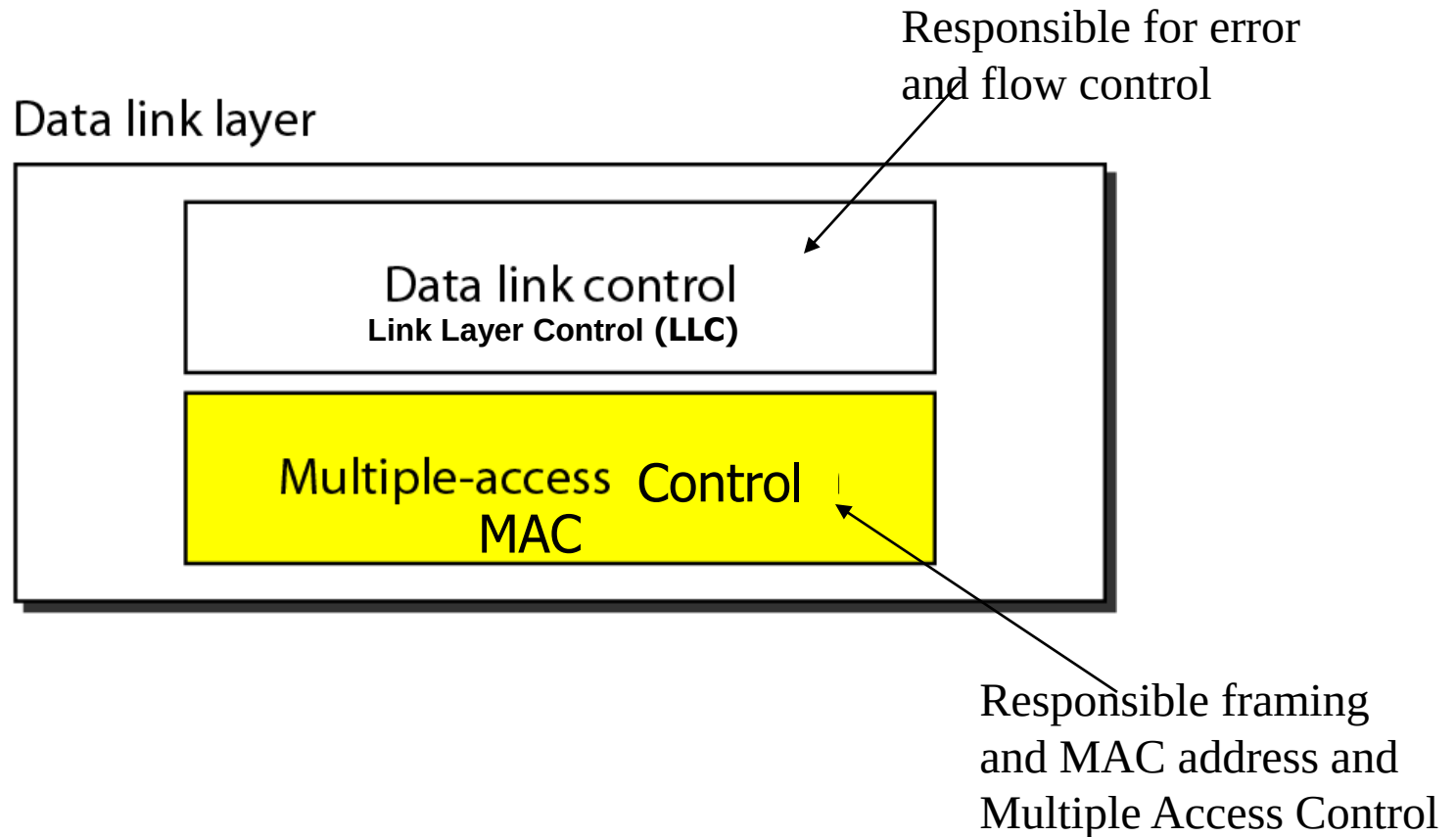


shared wireless
(e.g. Wavelan)



satellite

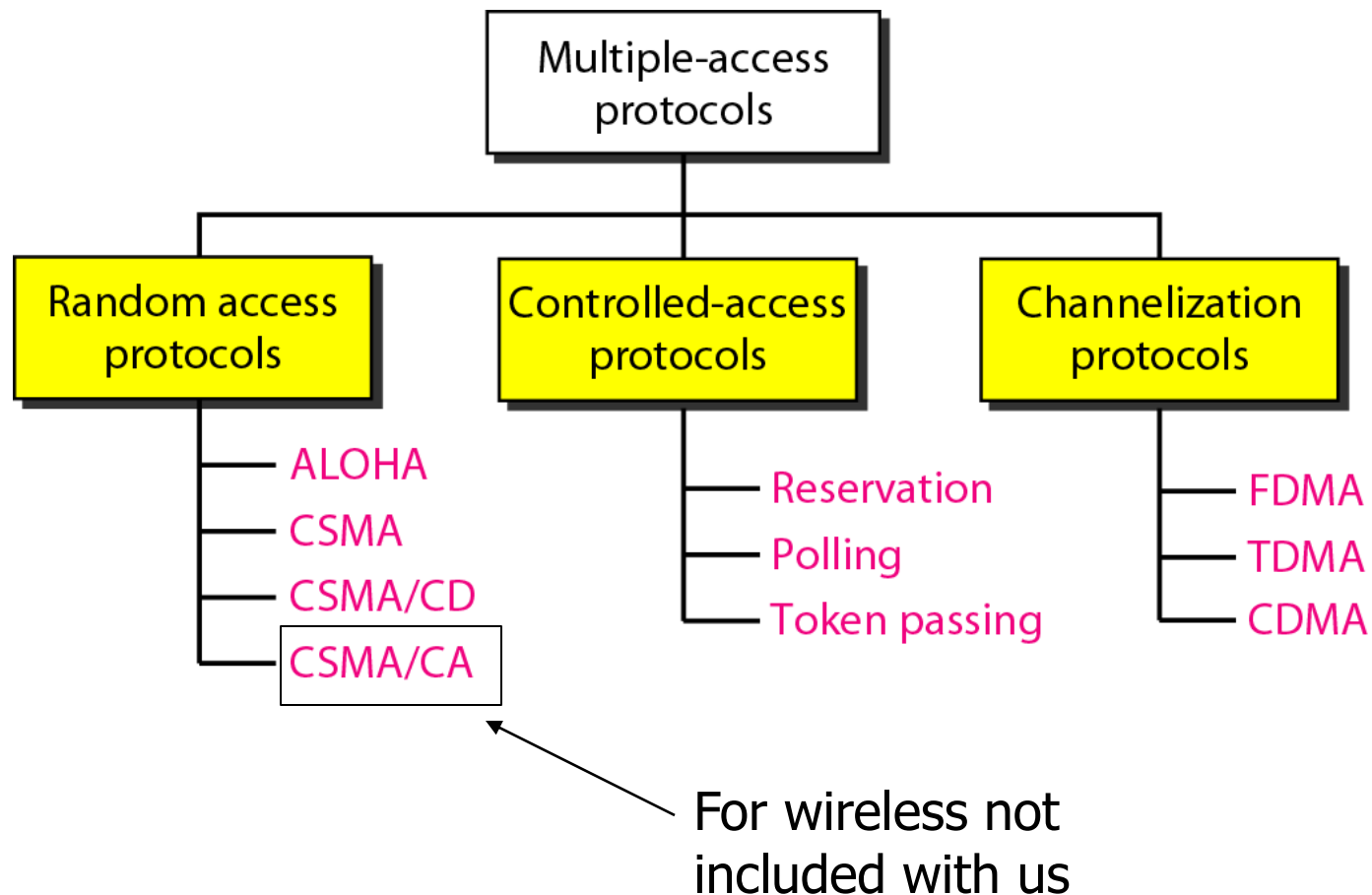
Figure 12.1 *Data link layer divided into two functionality-oriented sublayers*



Multiple Access

- **Problem:** When two or more nodes transmit at the same time, their frames will collide and the link bandwidth is **wasted** during collision
 - How to coordinate the access of multiple sending/receiving nodes to the shared link???
- **Solution:** We need a **protocol** to coordinate the transmission of the active nodes
- These protocols are called **Medium or Multiple Access Control (MAC) Protocols** belong to a **sublayer** of the data link layer called **MAC** (Medium Access Control)
- What is expected from Multiple Access Protocols:
 - Main task is to **minimize collisions** in order to **utilize the bandwidth** by:
 - Determining **when** a station can use the link (medium)
 - **what** a station should do when the link is **busy**
 - **what** the station should do when it is involved in **collision**

Figure 12.2 *Taxonomy of multiple-access protocols discussed in this chapter*



Random Access

- **Random Access (or contention) Protocols:**
 - No station is superior to another station and none is assigned the control over another.
 - A station with a frame to be transmitted **can use the link directly based** on a procedure defined by the protocol to make a decision on whether or not to send.
- **ALOHA Protocols**
- Was designed for **wireless LAN** and can be used for **any shared medium**

Random Access

- **Pure ALOHA Protocol Description**
 - All frames from any station are of fixed length (**L bits**)
 - Stations transmit at equal **transmission time** (*all stations produce frames with equal frame lengths*).
 - A station that has data **can transmit at any time**
 - **After transmitting a frame**, the sender **waits** for an **acknowledgment** for an amount of time (time out) equal to the **maximum round-trip propagation delay** = $2 * t_{\text{prop}}$ (**see next slide**)
 - If **no ACK** was received, sender assumes that the **frame or ACK** has been destroyed and **resends** that frame after it **waits for a random amount of time**
 - If station fails to receive an ACK after repeated transmissions, **it gives up**
 - **Channel utilization or efficiency or Throughput** is the **percentage** of the transmitted frames that arrive **successfully** (without collisions) **or** the **percentage** of the **channel bandwidth** that will be used for transmitting frames without collisions
 - ALOHA Maximum channel utilization is **18%** (i.e, if the system produces **F frames/s**, then **$0.18 * F$ frames will arrive successfully on average without the need of retransmission**).

Maximum Propagation Delay

- **Maximum propagation delay**(t_{prop}): time it takes for a bit of a frame to travel between the **two most widely** separated stations.

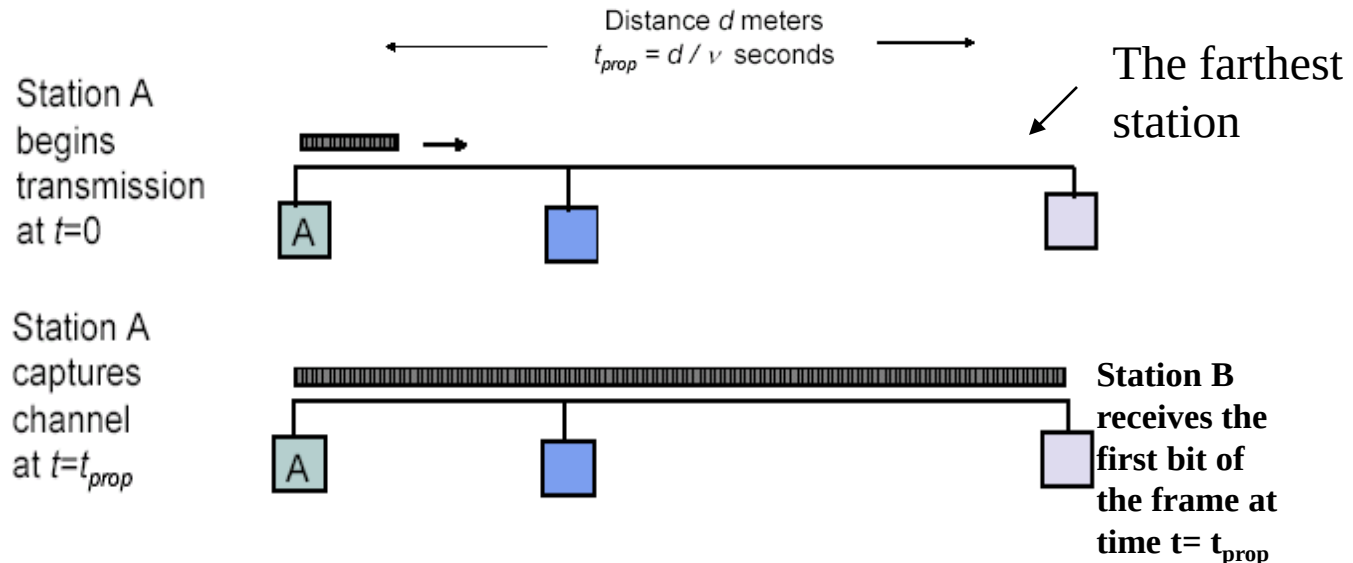
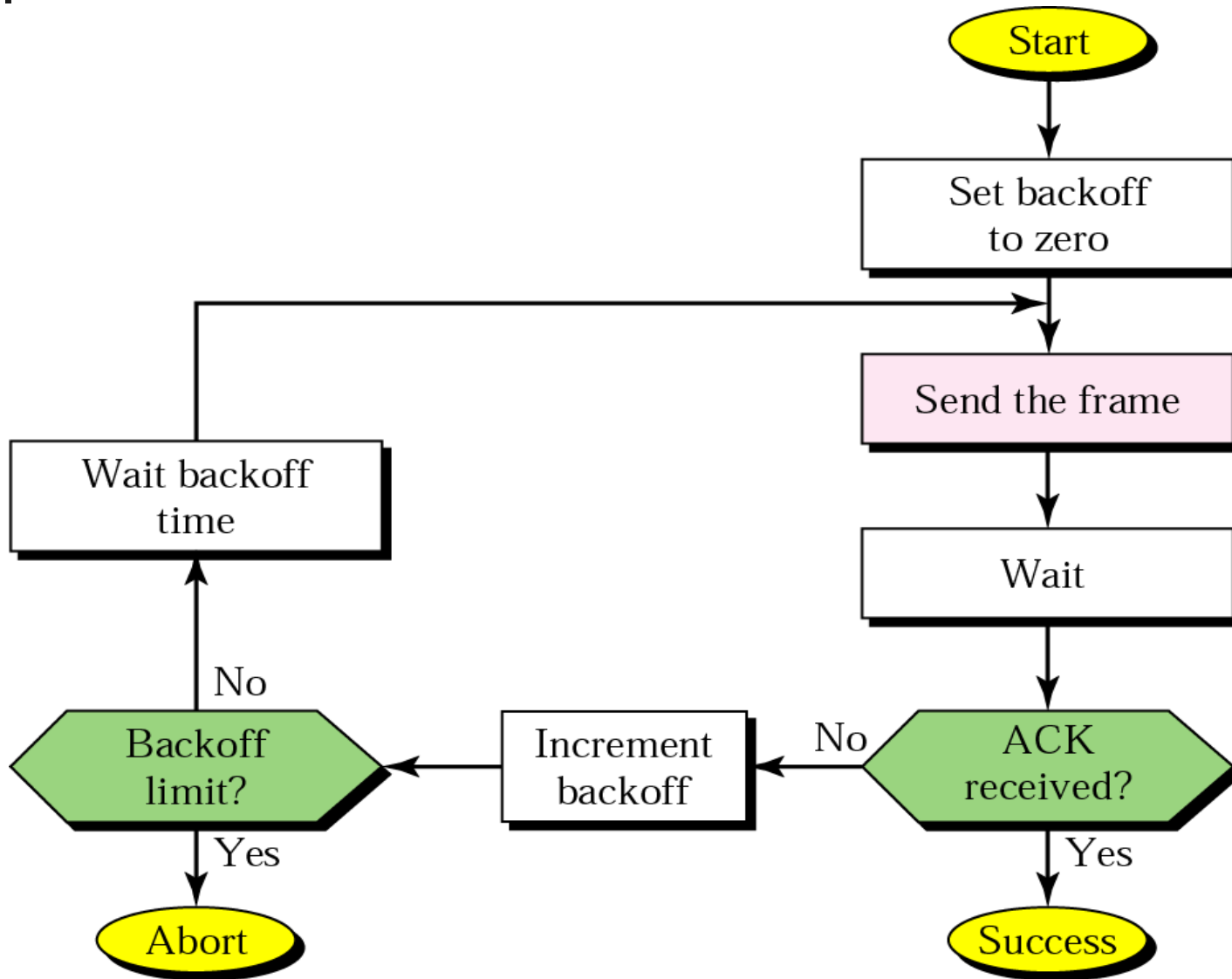
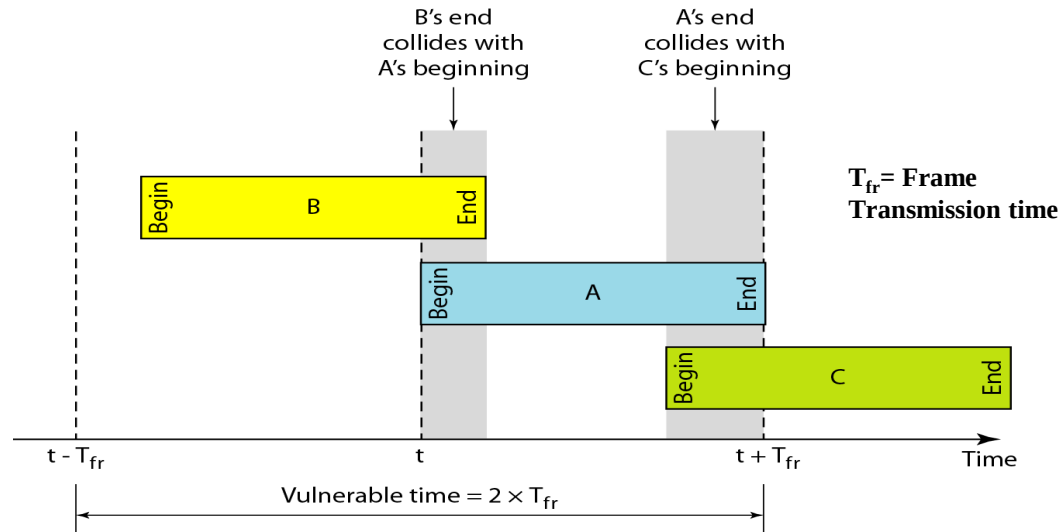


Figure 13.4 Procedure for ALOHA protocol



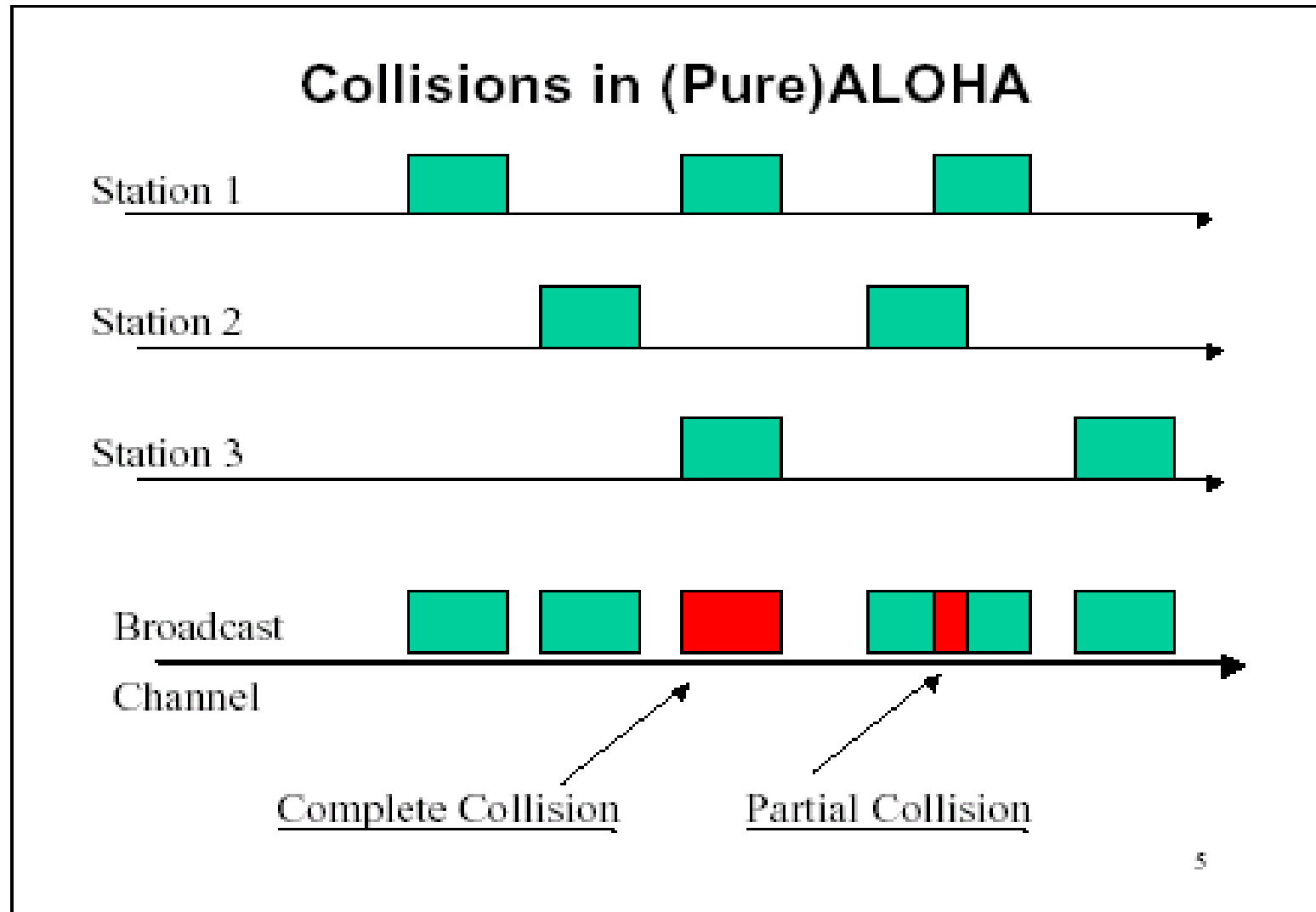
Critical time for pure ALOHA protocol



If the frame transmission time is T sec, then the vulnerable time is $= 2 T$ sec.

This means no station should send during the T -sec before this station starts transmission and no station should start sending during the T -sec period that the current station is sending.

Pure ALOHA



In pure ALOHA, frames are transmitted at completely arbitrary times.

Note

The throughput (S) for pure ALOHA is

$$S = G \times e^{-2G} .$$

The maximum throughput

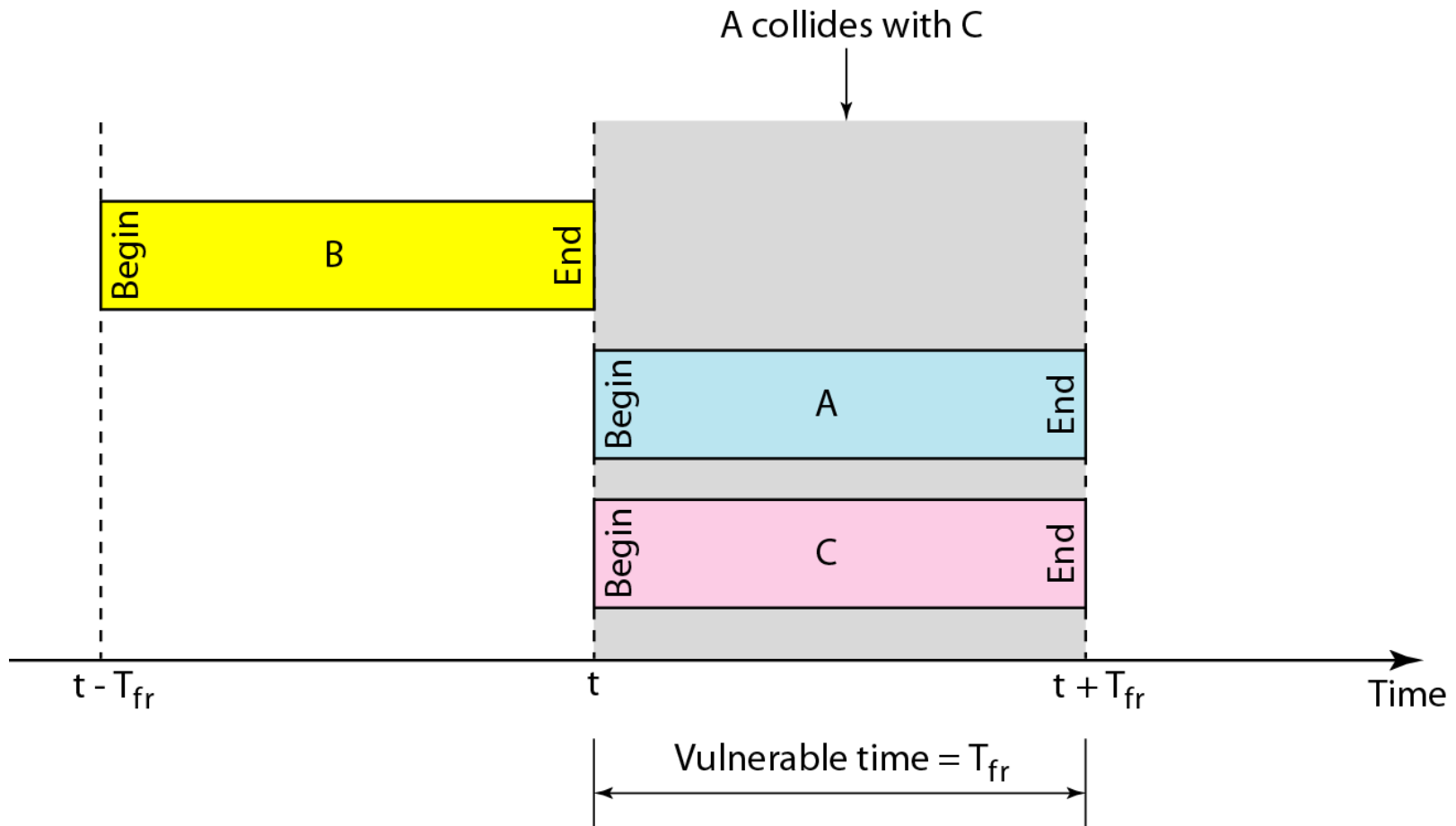
$$S_{\max} = 0.184 \text{ when } G = (1/2).$$

**G = Average number of frames generated by the system (all stations) during
one frame transmission time**

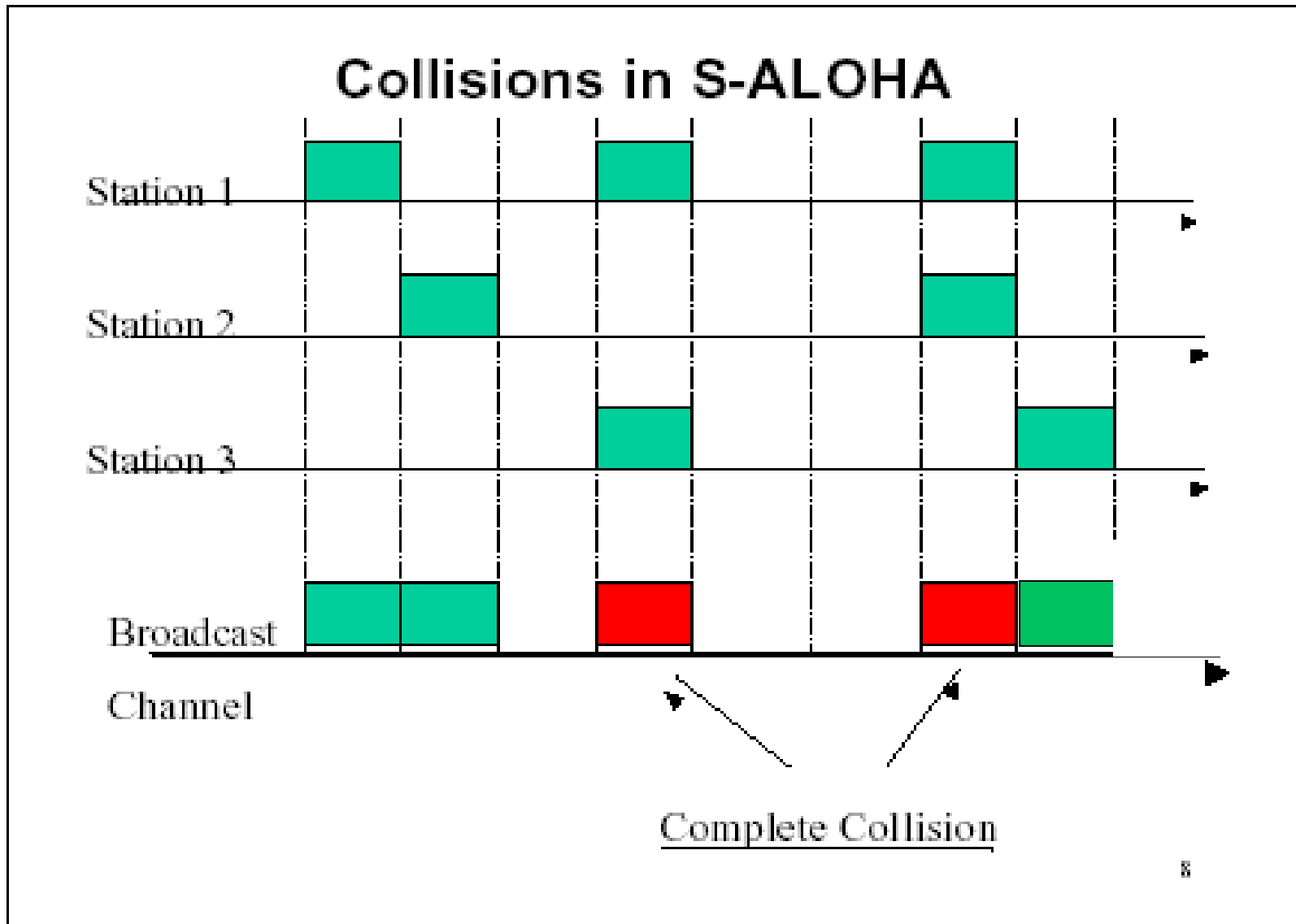
Random Access – Slotted ALOHA

- Time is divided into slots equal to a **frame transmission time (T_{fr})**
- A station can transmit at the beginning of a slot only
- If a station misses the beginning of a slot, it has to wait until the beginning of the next time slot.
- **A central clock** or station informs all stations about the start of a each slot
- Maximum channel utilization is **37%**

In danger time for slotted ALOHA protocol



Random Access – Slotted ALOHA





Note

The throughput for slotted ALOHA is

$$S = G \times e^{-G} .$$

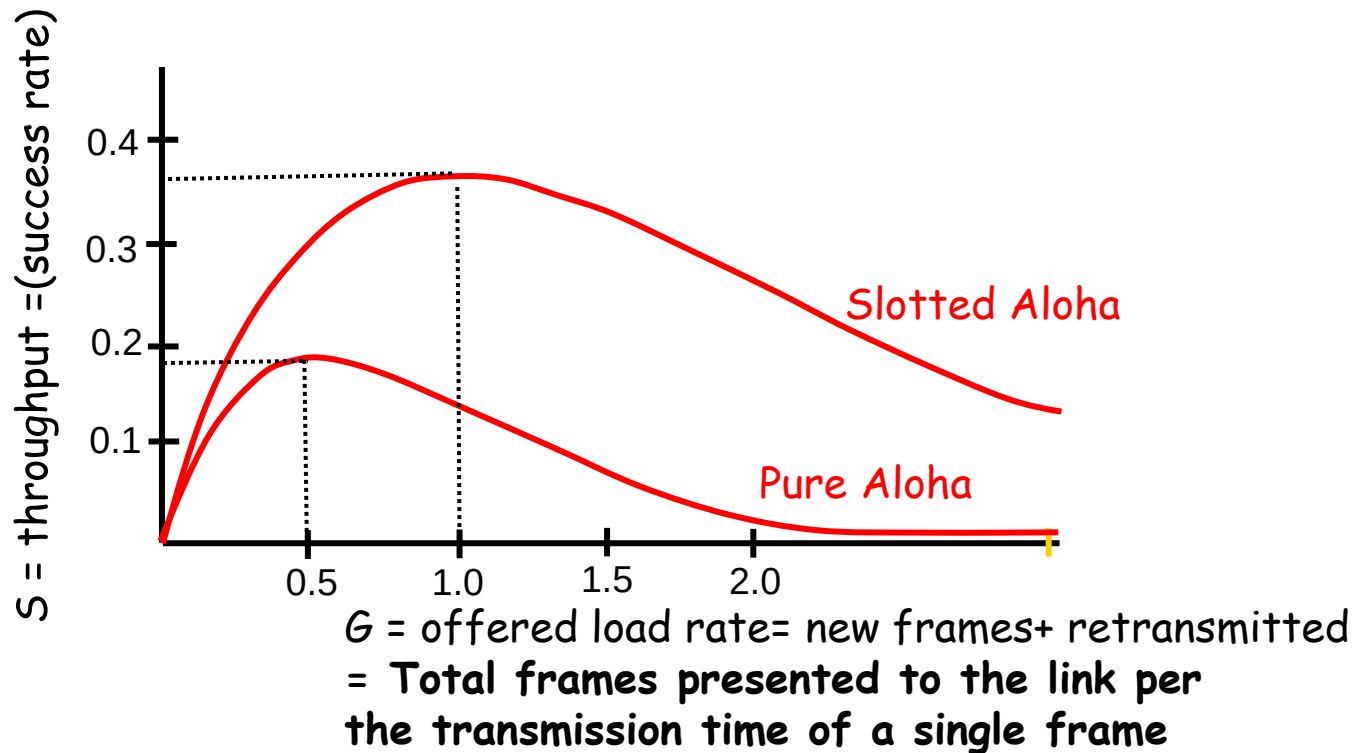
The maximum throughput

$$S_{\max} = 0.368 \text{ when } G = 1.$$

Differences between Pure ALOHA and slotted ALOHA

Pure ALOHA	Slotted ALOHA
Any Station can transmit data at any time.	Any station can transmit the data only at the beginning of any time slot.
Vulnerability time = 2 x Transmission time.	Vulnerability time = Transmission time.
The probability of successful transmission in Pure ALOHA is $G \times e^{-2G}$	The probability of successful transmission in Slotted ALOHA is $G \times e^{-G}$
The max probability of successful transmission in pure aloha occurs at $G=\frac{1}{2}$ and the value is $\frac{1}{2} \times e^{-1} = 18.39\%$	The max probability of successful transmission in slotted aloha occurs at $G=1$, the value is $1 \times e^{-1} = 36.79\%$
Time is continuous and not synchronised.	Time is discrete and Synchronised.

Efficiency of Aloha



■ Advantage of ALOHA protocols

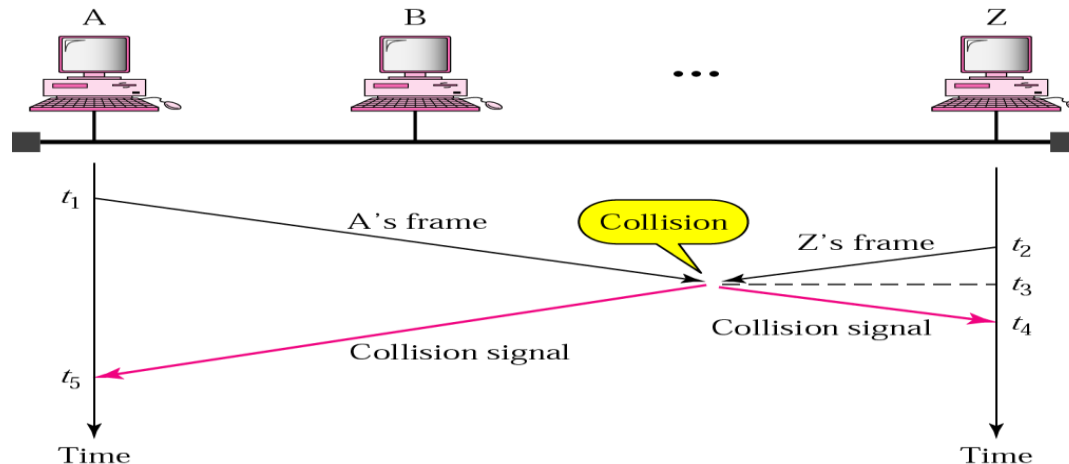
- A node that has frames to be transmitted can **transmit continuously** at the **full rate of channel (R bps)** if it is the only node with frames
- Simple to be implemented
- No master station is needed to control the medium

■ Disadvantage

- If (M) nodes want to transmit, many collisions can occur and the rate allocated for each node will **not be on average R/M bps**
- This causes low channel utilization

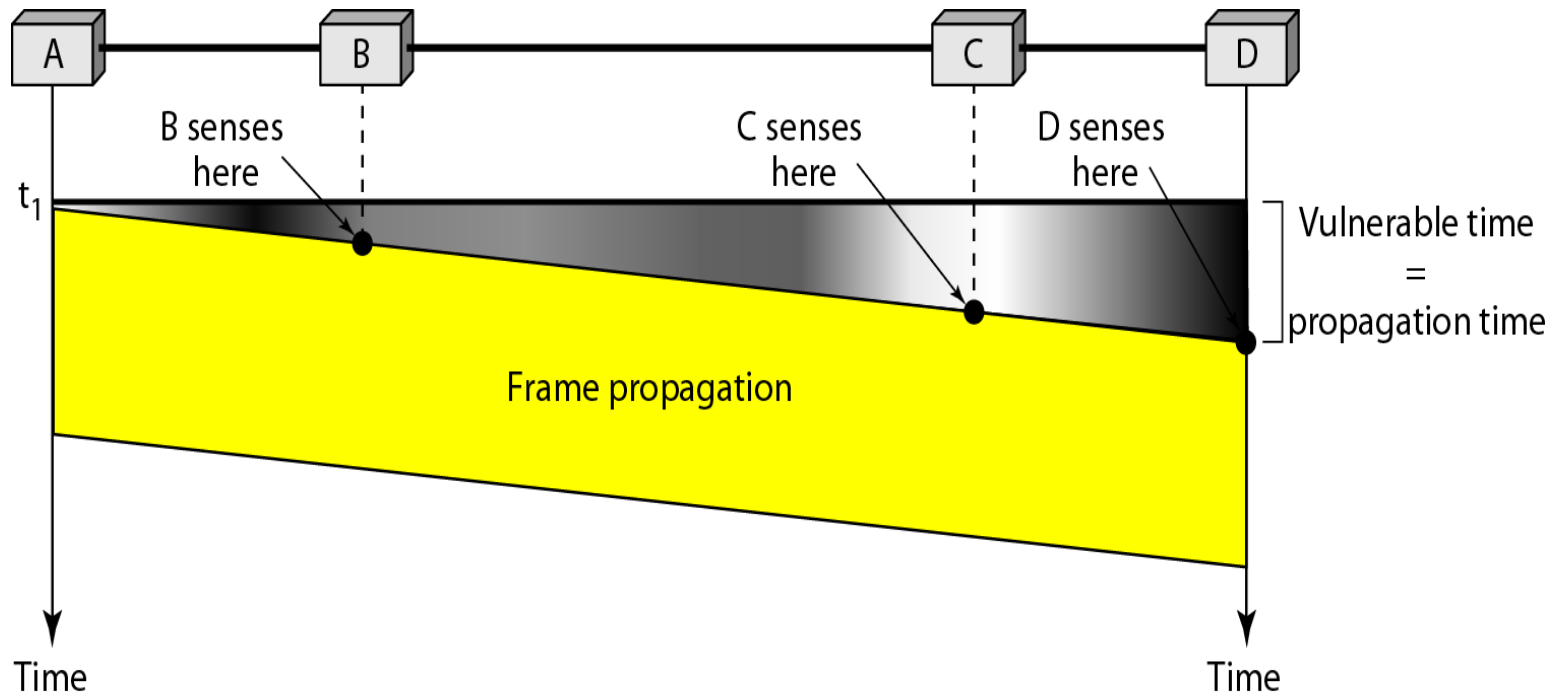
Random Access – Carrier Sense Multiple Access (CSMA)

- To improve performance, avoid transmissions that are certain to cause collisions
- Based on the fact that in LAN propagation time is **very small**
- ➔ If a frame was sent by a station, All stations knows immediately so they can **wait before start sending**
 - ➔ A station with frames to be sent, should **sense the medium** for the presence of another transmission (carrier) before it starts its own transmission
- This can **reduce** the possibility of collision but it cannot eliminate it.
 - Collision can only happen when more than one station begin transmitting within a short time (the **propagation time** period)



Random Access – Carrier Sense Multiple Access (CSMA)

- Vulnerable time for CSMA is the **maximum propagation time**
- The longer the propagation delay, the worse the performance of the protocol because of the above case.



Types of CSMA Protocols

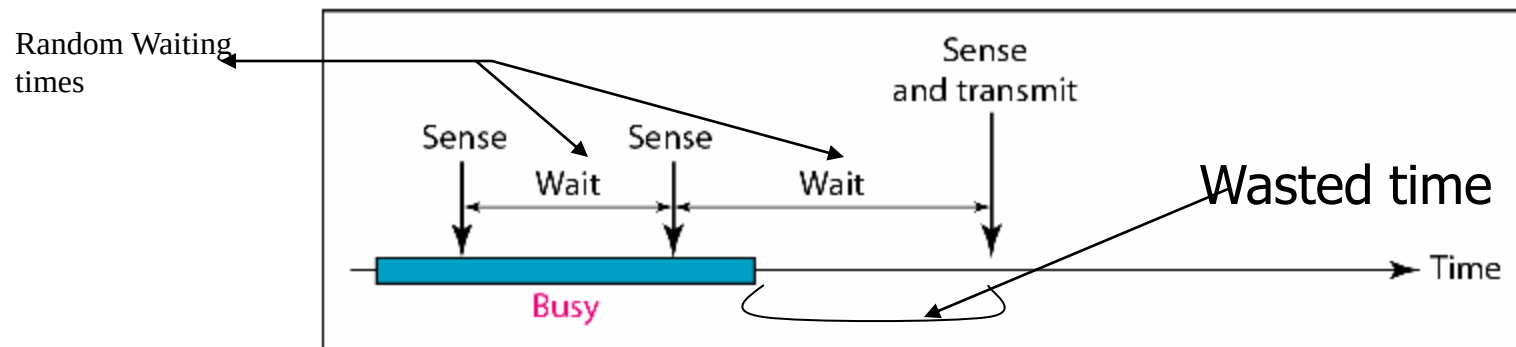
Different CSMA protocols that determine:

- What a station should do when the medium is **idle**?
- What a station should do when the medium is **busy**?

1. Non-Persistent CSMA
2. 1-Persistent CSMA
3. p-Persistent CSMA

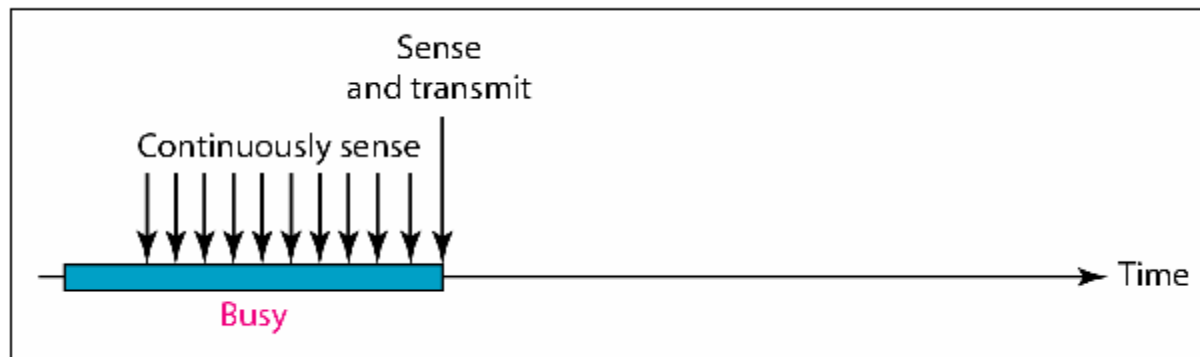
Nonpersistent CSMA

- A station with frames to be sent, should sense the medium
 1. If medium is idle, **transmit**; otherwise, go to 2
 2. If medium is busy, (**backoff**) wait a **random amount of time** and repeat 1
- Non-persistent Stations are **deferential (respect others)**
- Performance:
 - Random delays reduces probability of collisions because two stations with data to be transmitted will wait for different amount of times.
 - Bandwidth is **wasted** if waiting time (backoff) is large because medium will remain idle following end of transmission even if one or more stations have frames to send



1-persistent CSMA

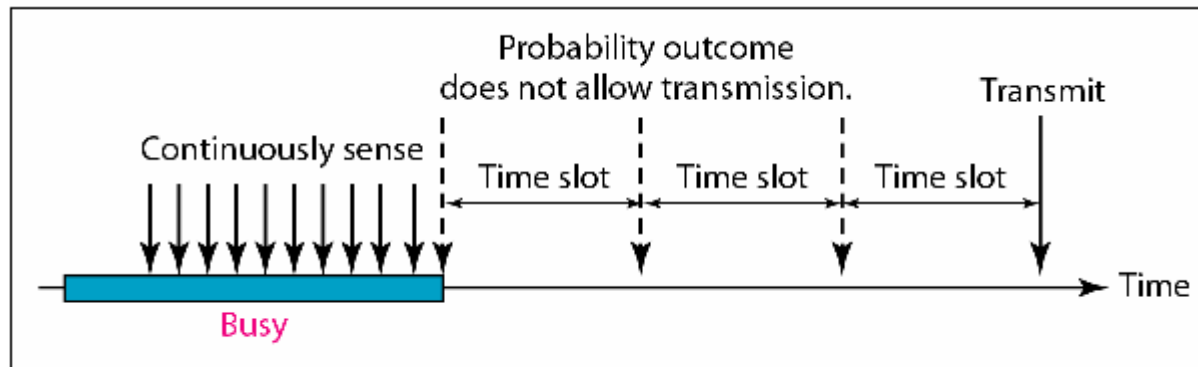
- To avoid idle channel time, 1-persistent protocol used
- Station wishing to transmit listens to the medium:
 1. If medium idle, **transmit** immediately;
 2. If medium busy, **continuously listen** until medium becomes idle; then transmit immediately with probability 1
- Performance
 - 1-persistent stations are **selfish**
 - If two or more stations becomes ready at the same time, **collision guaranteed**



a. 1-persistent

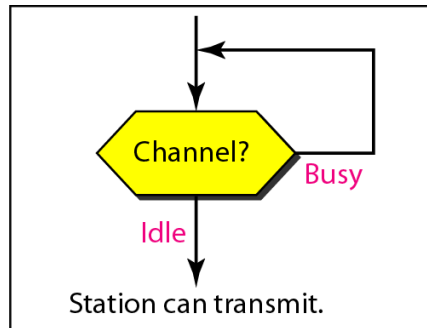
P-persistent CSMA

- Time is divided to slots where each Time unit (slot) typically equals **maximum propagation delay**
- Station wishing to transmit listens to the medium:
 1. If medium idle,
 - transmit with probability (p), OR
 - wait **one time unit (slot)** with probability ($1 - p$), then repeat 1.
 2. If medium busy, **continuously listen until idle** and repeat step 1
 3. Performance
 - Reduces the possibility of collisions like **nonpersistent**
 - Reduces channel idle time like **1-persistent**

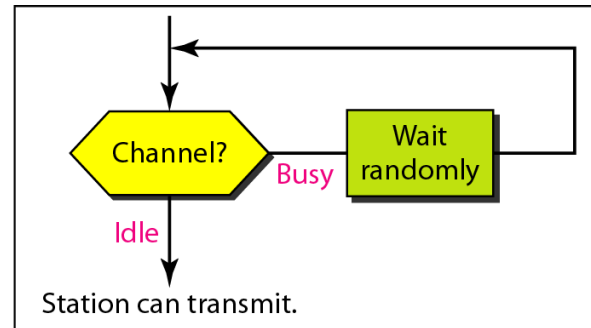


c. p-persistent

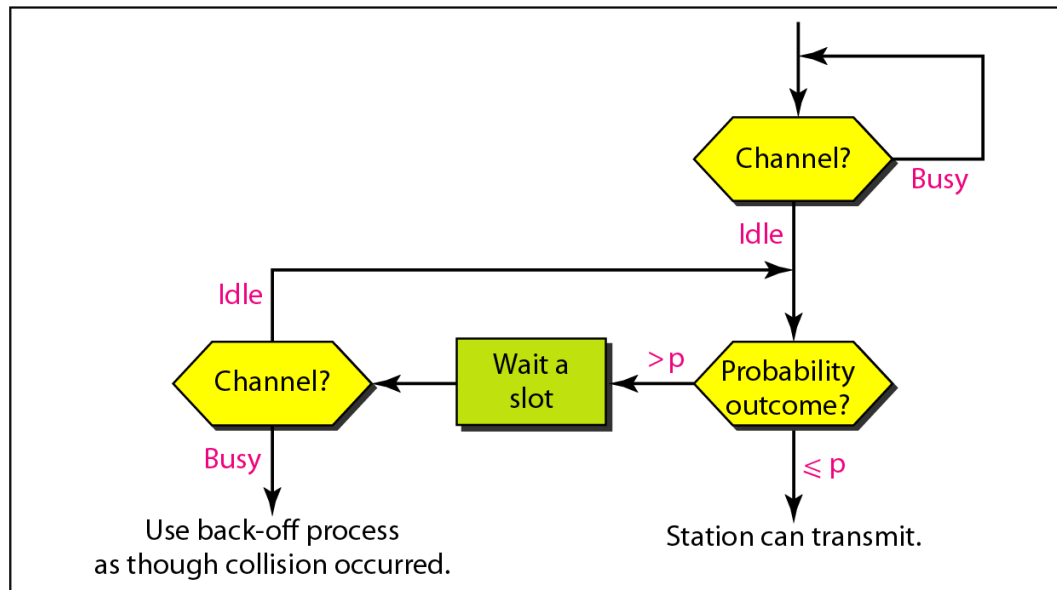
Flow diagram for three persistence methods



a. 1-persistent

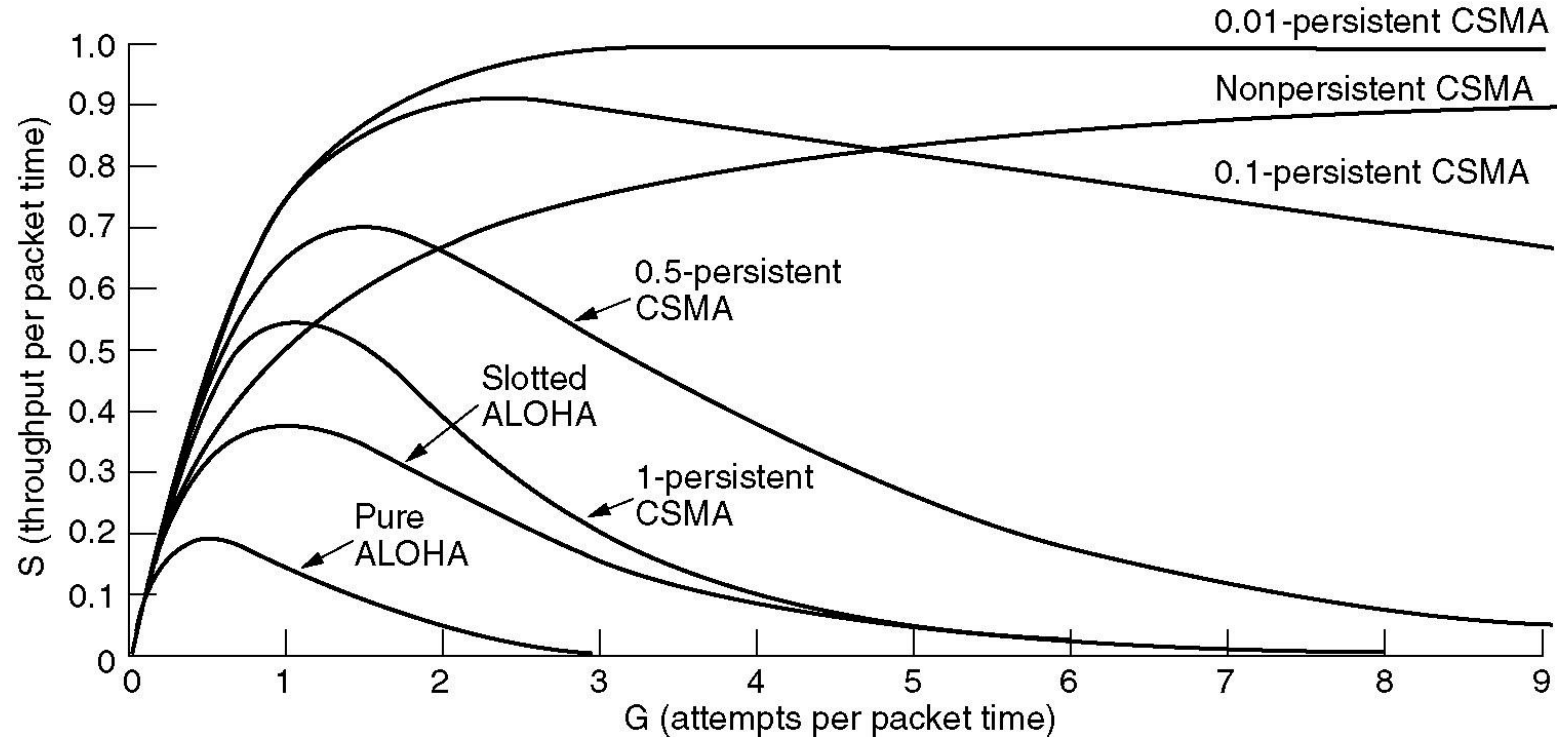


b. Nonpersistent



c. p-persistent

Persistent and Nonpersistent CSMA



Comparison of the channel utilization versus load for various random access protocols.

CSMA/CD (Collision Detection)

- *CSMA (all previous methods) has an inefficiency:*
 - If a collision has occurred, the channel is **unstable** until colliding packets have **been fully transmitted**
- *CSMA/CD (Carrier Sense Multiple Access with Collision Detection) overcomes this as follows:*
 - While transmitting, the sender is **listening to medium** for collisions.
 - Sender **stops transmission** if collision has occurred **reducing channel wastage** .

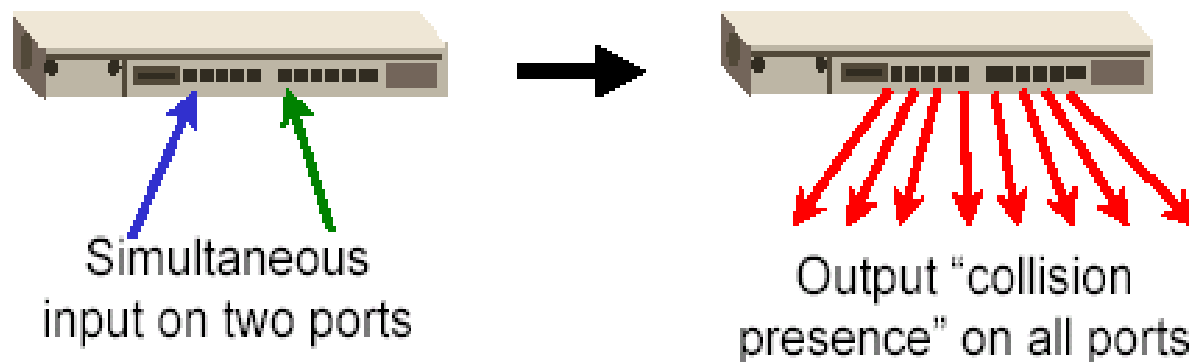
CSMA/CD is Widely used for **bus topology LANs** (IEEE 802.3, **Ethernet**).

How does a node detect a collision?

Transceiver: A node monitors the media while transmitting. If the observed power is more than transmitted power of its own signal, it means collision occurred



Hub: if input occurs simultaneously on two ports, it indicates a collision. Hub sends a collision presence signal on all ports.



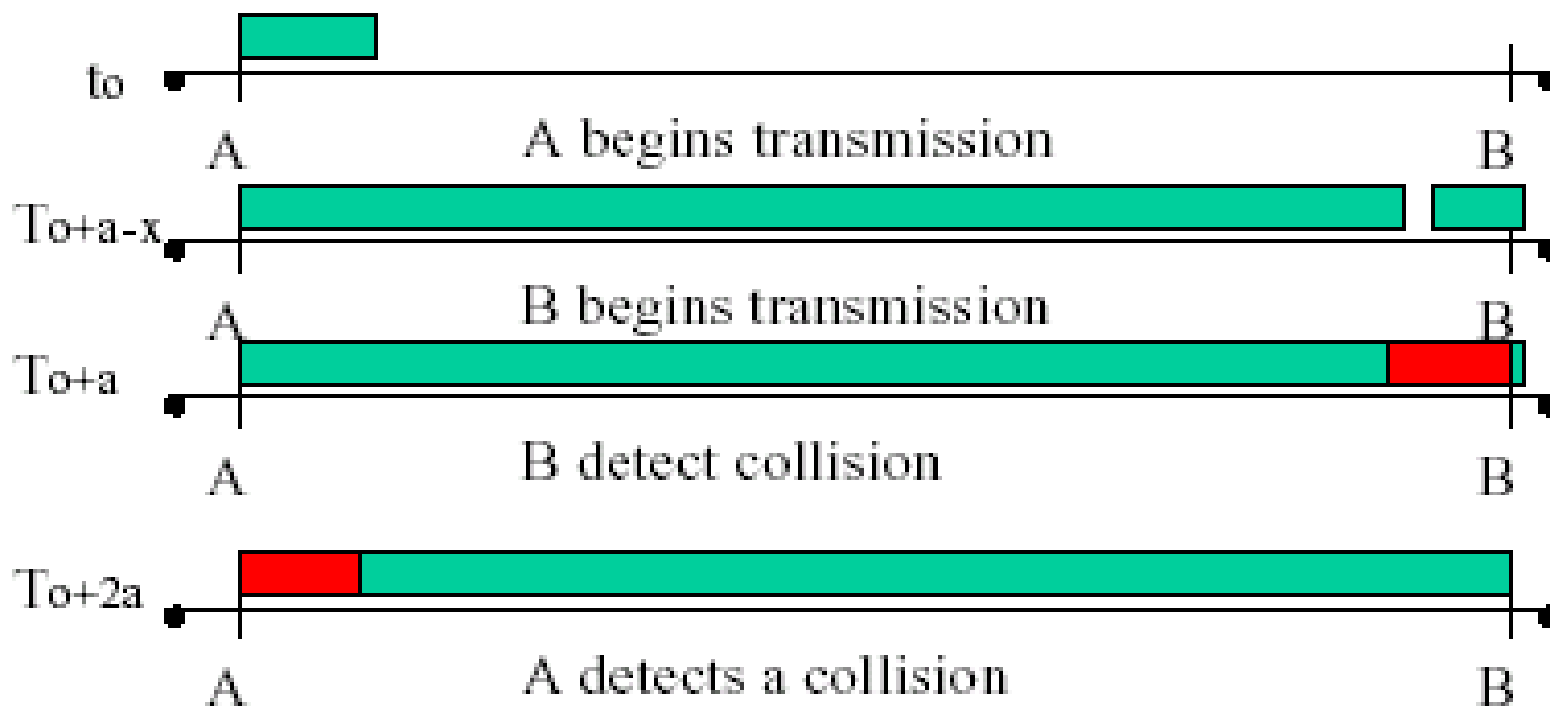
CSMA/CD Protocol

- Use one of the CSMA persistence algorithm (*non-persistent, 1-persistent, p-persistent*) for transmission
- If a collision is detected by a station during its transmission then it should do the following:
 - **Abort transmission** and
 - **Transmit a *jam signal* (48 bit)** to notify other stations of collision so that they will **discard the transmitted frame** also to make sure that the collision signal will stay until detected by the furthest station
 - After sending the *jam signal*, **backoff (wait) for a *random*** amount of time, then
 - Transmit the frame again

CSMA/CD

- **Question:** How long does it take to detect a collision?
- **Answer:** In the *worst case*, **twice the maximum propagation delay of the medium**

Note: a = maximum propagation delay



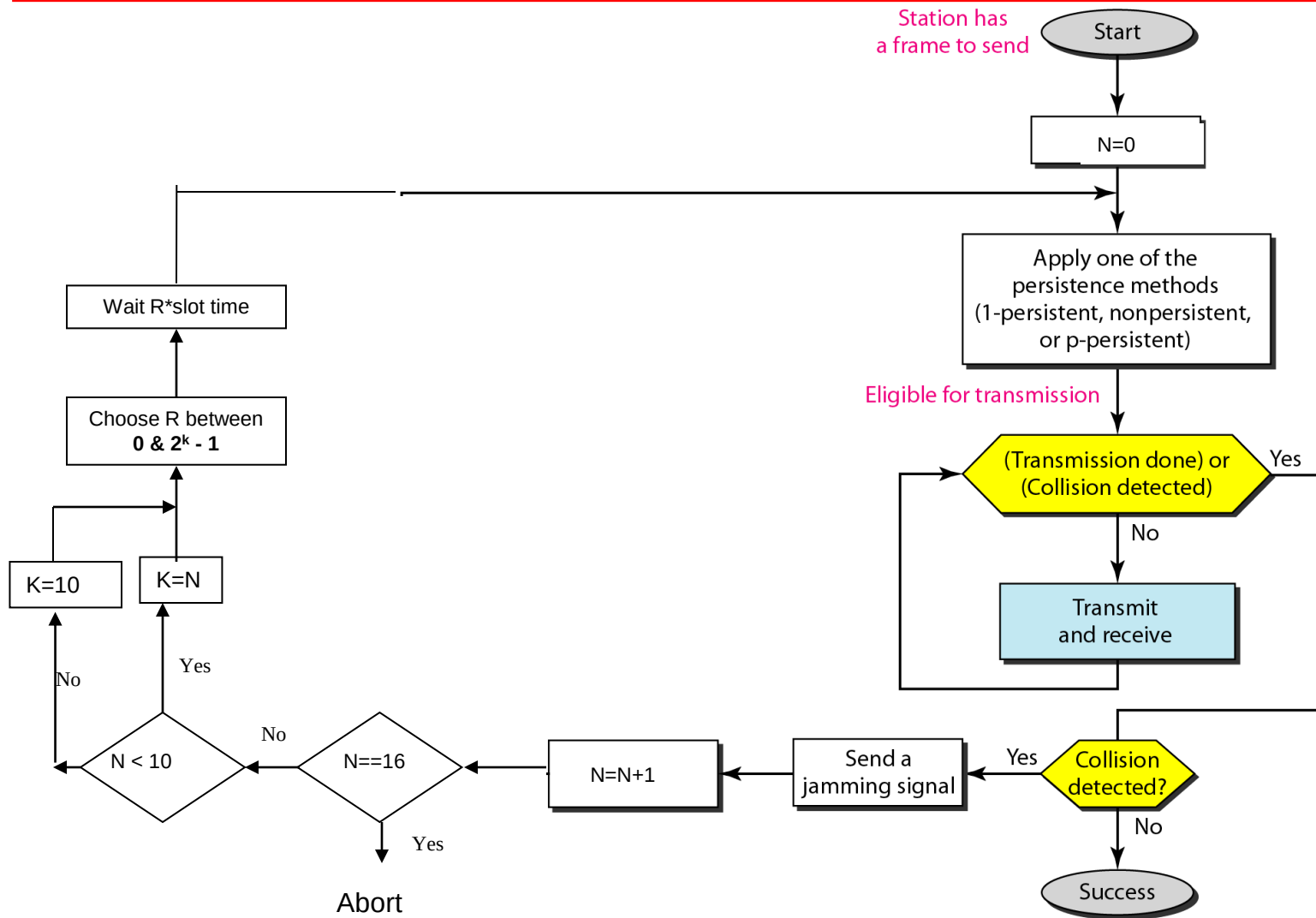
CSMA/CD

- Restrictions of CSMA / CD:
 - Packet **transmission time** should be **at least** as long as the time needed to detect a collision ($2 * \text{maximum propagation delay} + \text{jam sequence transmission time}$)
 - Otherwise, CSMA/CD does not have an advantage over CSMA

Exponential Backoff Algorithm

- Ethernet uses the **exponential backoff algorithms** to determine **the best duration of the random waiting period after the collision happens**
- **Algorithm:**
 - Set “**slot time**” equal to $2 \times \text{maximum propagation delay} + \text{Jam sequence transmission time}$ (= 51.2 usec for Ethernet **10-Mbps** LAN)
 - After K^{th} collision, select a random number (R) between 0 and $2^k - 1$ and **wait** for a period equal to $(R \times \text{slot time})$ then **retransmit** when the medium is **idle, for example:**
 - After first collision ($K=1$), select a number (R) between 0 and $2^1 - 1$ {0,1} and wait for a period equal to $R \times \text{slot times}$ (Wait for a period 0 usec or 1×51.2 usec) then retransmit when the medium is idle
 - Do not increase random number range, if $K=10$
 - ➔ Maximum interval {0 – 1023}
 - Give up after 16 unsuccessful attempts and report failure to higher layers

Figure 12.14 *Flow diagram for the CSMA/CD*



Exponential Backoff Algorithm

- Reduces the chance of two waiting stations picking the same random waiting time
- When network traffic is light, it results in **minimum** waiting time before transmission
- As congestion increases (traffic is high), collisions increase, stations backoff by **larger amounts** to reduce the probability of collision.
- Exponential Back off algorithm gives **last-in, first-out effect**
 - Stations with **no or few collisions** will have the chance to transmit before stations that have waited longer because of their previous unsuccessful transmission attempts.

Performance of Random Access Protocols

- Simple and easy to implement
- Decentralized (no central device that can fail and bring down the entire system)
- In low-traffic, packet transfer has low-delay
- However, limited throughput and in heavier traffic, packet delay has no limit.
- In some cases, a station may never have a chance to transfer its packet. (**unfair protocol**)
- A node that has frames to be transmitted can **transmit continuously** at the **full rate of channel (R)** if it is the **only node with frames**
- If (M) nodes want to transmit, many collisions can occur and the rate for each node will **not be on average R/M**